1. ***This is the start of a multipart question. The information below pertains to question 1 to question 8.***

   The data set `card` contains data on young men in the US from the 1976 National Longitudinal Survey of Young Men. You can load the data into R using the `wooldridge` library using:

   ```
   library(wooldridge)
   dat = wooldridge::card
   ```

   The main outcome of interest is the variable *lwage*, which is the log of the hourly wage. What is the mean of the variable *lwage* in the dataset?

   a. 6.2004
   b. 6.3520
   c. 6.1395
   d. 5.9810
   e. None of the above

2. How many missing values or NA's are there for the variable *IQ* in the dataset?

   a. 47
   b. 949
   c. 690
   d. 0
   e. None of the above

3. Consider the following linear regression model.

$$\begin{aligned} lwage = \beta_0 + \beta_1(educ) + \beta_2(exper) + \beta_3(exper^2) + \beta_4(black) + \\ \beta_5(smsa) + \beta_6(south) + \beta_7(momdad14) + \beta_8(KWW) + \varepsilon \end{aligned}$$

   The model includes controls for education, experience and experience squared, and a test score on knowledge of the world of work (*KWW*). Finally, (*black, smsa, south, momdad14*), indicator variables for race, living in a metropolitan area, living in the south, and living with both parents at age 14 are included in the model.
   Estimate the model above with OLS. What is the estimated coefficient on education?

   a. 0.05482
   b. 0.06213
   c. 0.05570
   d. 0.04911
   e. 0.05823

4. What is the sum of squared errors of the model?

   a. 392.3281
   b. 405.3401

c. 0.3704
d. 410.1124
e. None of the above

5. Suppose we wanted to create a confidence interval for one of the parameters of the model. If we wanted to create a 90% confidence interval, what critical value would we use?

   a. 1.6449
   b. 1.2818
   c. 1.9608
   d. 1.6454
   e. None of the above

6. Consider the null hypothesis that $H_0 : \beta_{momdad14} = 0$. What is the strongest conclusion we can draw about this null hypothesis?

   a. Reject the null at the 0.1% level
   b. Reject the null at the 1% level
   c. Reject the null at the 5% level
   d. Reject the null at the 10% level
   e. Fail to reject the null at the 10% level

7. Consider the null hypothesis that $H_0 : \beta_{educ} = 0.05$. What is the p-value associated with this hypothesis test?

   a. 0.0925
   b. 0.8150
   c. 0.2533
   d. 0.1850
   e. None of the above

8. What is the covariance between the estimate of the parameter on *educ* and the parameter on *KWW*?

   a. 9.3490×10^{-6}
   b. -2.7081×10^{-6}
   c. 2.7081×10^{-6}
   d. -2.6182×10^{-6}
   e. -1.2120×10^{-5}

9. ***This is the start of a multipart question. The information below pertains to question 9 to question 15.***

   The data set `mroz` contains data on married women in the US from the 1975 Panel Study of Income Dynamics. You can load the data into R using the `wooldridge` library using:

```
library(wooldridge)
dat = wooldridge::mroz
```

We would like to study what are the main determinants of earning a high wage using a linear probability model. To answer this question you will need to add a new variable to the data set called *highwage* which should equal 1 if the woman's hourly wage is above \$5 (i.e., *wage* > 5) and zero otherwise. Women who did not work have a missing wage, so *highwage* should also be missing for them.
Among women with a reported wage, what percent are *highwage*?

a. 13.94%
b. 75.47%
c. 24.53%
d. 56.84%
e. None of the above

10. Estimate a linear probability model on *highwage*.
Include in your model controls for years of education (*educ*), years of work experience (*exper*), *age*, and an indicator for living in a city (*city*). Your model should have 5 parameters including the intercept.
How many observations are used to estimate the model?

a. 753
b. 428
c. 325
d. 423
e. None of the above

11. What is the correct interpretation of the parameter on *educ*? Holding fixed the variables in the model, one additional year of education increases the probability of earning a high wage by ...

a. 7.7463 percent
b. 0.0775 percentage points
c. 0.7746 percentage points
d. 7.7463 percentage points
e. None of the above

12. What is the estimate of the lower bound of the 90% confidence interval for the intercept of the model?

a. –1.1335
b. –0.5772
c. –1.0847
d. –1.0613
e. None of the above

13. Consider the null hypothesis that $H_0 : \beta_{city} = 0$. What is the strongest conclusion we can draw about this null hypothesis?

    a. Reject the null at the 0.1% level
    b. Reject the null at the 1% level
    c. Reject the null at the 5% level
    d. Reject the null at the 10% level
    e. Fail to reject the null at the 10% level

14. Re-estimate the model adding husband's years of education (*huseduc*) as an additional control. What is the percentage change in the parameter on *educ* when we control for husband's education?

    a. –8.63%
    b. 8.63%
    c. 7.95%
    d. 108.63%
    e. None of the above

15. Re-estimate the model adding husband's years of education (*huseduc*) as an additional control. What is the variance inflation factor for the parameter on *educ* when we control for husband's education? (note: if you are calculating the variance inflation factor from the formula you need to ensure that the same observations are used in the different regressions)

    a. 1.6279
    b. 1.6137
    c. 0.6441
    d. 1.5525
    e. None of the above

Solution Key

Q	Ans	Value	How / trap
1	**e**	6.2618	`mean(dat$lwage)`; not listed
2	**b**	949	`sum(is.na(dat$IQ))`; 47 = *KWW*, 690 = *fatheduc*
3	**c**	0.05570	lm drops the 47 rows with missing *KWW*, $n = 2963$
4	**b**	405.3401	`deviance(m)`; 0.3704 is $\hat{\sigma}$
5	**d**	1.6454	`qt(0.95, df.residual(m))`, df = 2954; 1.6449 is `qnorm`, 1.2818 uses 0.90
6	**e**	$p = 0.1134$	$p > 0.10$
7	**d**	0.1850	`linearHypothesis(m, "educ = 0.05"), Pr(>F)`; 0.0925 is one-sided
8	**b**	-2.71×10^{-6}	`vcov(m)["educ","KWW"]`; 9.35×10^{-6} is (*educ*, *exper*)
9	**c**	24.53%	`mean(dat$highwage, na.rm=TRUE)`; 13.94% divides by all 753 women
10	**b**	428	`nobs(m1)`: the 325 non-workers have NA outcome and are dropped
11	**d**	0.0775	LPM: $\times 100$ = percentage points
12	**c**	–1.0847	`confint(m1, "(Intercept)", level=0.90)`; –1.1335 is the 95% bound
13	**e**	$p = 0.2229$	$p > 0.10$
14	**b**	8.63%	`(new - old) / old * 100`
15	**d**	1.5525	`car::vif(m2)`. By hand on all 753 rows (wrong sample) gives 1.6279; 1.6137 is the VIF of *huseduc*

R code

```
# ECON 5021 Unit 2 Data Analysis Practice C: 26101200004 - solution key
# Key: 1e 2b 3c 4b 5d 6e 7d 8b 9c 10b 11d 12c 13e 14b 15d

library(wooldridge)
library(car)

# ---- Q1-8: card ----
dat = wooldridge::card

# Q1. What is the mean of the variable lwage in the dataset?
mean(dat$lwage)                                  # 6.2618  (e)

# Q2. How many missing values or NA's are there for the variable IQ?
sum(is.na(dat$IQ))                               # 949     (b)

# Q3. Estimate the model with OLS. Coefficient on educ?
m = lm(lwage ~ educ + exper + I(exper^2) + black + smsa + south + momdad14 + KWW, data = dat)
summary(m)                                       # 0.05570 (c)   n = 2963 (KWW has 47 NAs)
```

```
# Q4. What is the sum of squared errors of the model?
deviance(m)                                  # 405.3401 (b)

# Q5. Critical value for a 90% confidence interval?
qt(0.95, df.residual(m))                     # 1.6454  (d)   qnorm gives 1.6449

# Q6. H0: b_momdad14 = 0. Strongest conclusion?
summary(m)$coefficients["momdad14", ]        # p = 0.1134 -> fail to reject at 10% (e)

# Q7. H0: b_educ = 0.05. p-value?
linearHypothesis(m, "educ = 0.05")           # Pr(>F) 0.1850  (d)   F = t^2, same p as the t
    -test

# Q8. Covariance between the estimates on educ and KWW?
vcov(m)["educ", "KWW"]                       # -2.7081e-06 (b)

# ---- Q9-15: mroz ----
dat = wooldridge::mroz
dat$highwage = as.numeric(dat$wage > 5)      # NA whenever wage is NA (women not working)

# Q9. Among women with a reported wage, what percent are highwage?
100 * mean(dat$highwage, na.rm = TRUE)       # 24.53%  (c)   NOT sum/753 = 13.94%

# Q10. LPM of highwage on educ, exper, age, city. How many observations are used?
m1 = lm(highwage ~ educ + exper + age + city, data = dat)
nobs(m1)                                     # 428     (b)

# Q11. Interpretation of the parameter on educ?
coef(m1)["educ"]                             # 0.077463 -> 7.7463 percentage points (d)

# Q12. Lower bound of the 90% CI for the intercept?
confint(m1, "(Intercept)", level = 0.90)     # -1.0847 (c)

# Q13. H0: b_city = 0. Strongest conclusion?
summary(m1)$coefficients["city", ]           # p = 0.2229 -> fail to reject at 10% (e)

# Q14. Add huseduc. Percentage change in the parameter on educ?
m2 = lm(highwage ~ educ + exper + age + city + huseduc, data = dat)
old = coef(m1)["educ"]
new = coef(m2)["educ"]
(new - old) / old * 100                      # 8.63%   (b)

# Q15. VIF for educ when we control for huseduc?
vif(m2)["educ"]                              # 1.5525  (d)
# by hand: same observations as m2 (drop the rows where highwage is NA)
d = dat[!is.na(dat$highwage), ]
1 / (1 - summary(lm(educ ~ exper + age + city + huseduc, data = d))$r.squared)       # 1.5525
# WRONG: full sample (753 rows) gives 1.6279
```